This fact, together with equation (5) and an application of Lemma 2.4 (with $a = x/n$), gives

$$\begin{aligned}\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{x^{s+1}}{s(s+1)}\left(-\frac{\zeta'(s)}{\zeta(s)}\right)ds &= x\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{(x/n)^s}{s(s+1)}\,ds\\ &= x\sum_{n\le x}\Lambda(n)\left(1-\frac{n}{x}\right)\\ &= \psi_1(x),\end{aligned}$$

as was to be shown.

2.1 Proof of the asymptotics for ψ_1

In this section, we will show that

$$\psi_1(x) \sim x^2/2 \quad \text{as } x \to \infty,$$

and as a consequence, we will have proved the prime number theorem.

The key ingredients in the argument are:

- the formula in Proposition 2.3 connecting ψ_1 to ζ, namely

  $$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{x^{s+1}}{s(s+1)}\left(-\frac{\zeta'(s)}{\zeta(s)}\right)ds$$

  for $c > 1$.

- the non-vanishing of the zeta function on $\mathrm{Re}(s) = 1$,

  $$\zeta(1+it) \neq 0 \quad \text{for all } t \in \mathbb{R},$$

  and the estimates for ζ near that line given in Proposition 2.7 of Chapter 6 together with Proposition 1.6 of this chapter.

Let us now discuss our strategy in more detail. In the integral above for $\psi_1(x)$ we want to change the line of integration $\mathrm{Re}(s) = c$ with $c > 1$, to $\mathrm{Re}(s) = 1$. If we could achieve that, the size of the factor x^{s+1} in the integrand would then be of order x^2 (which is close to what we want) instead of x^{c+1}, $c > 1$, which is much too large. However, there would still be two issues that must be dealt with. The first is the pole of $\zeta(s)$ at $s = 1$; it turns out that when it is taken into account, its contribution is exactly the main term $x^2/2$ of the asymptotic of $\psi_1(x)$. Second, what remains must be shown to be essentially smaller than this term, and so we must further refine the crude estimate of order x^2 when integrating on the line $\mathrm{Re}(s) = 1$. We carry out our plan as follows.

Fix $c > 1$, say $c = 2$, and assume x is also fixed for the moment with $x \geq 2$. Let $F(s)$ denote the integrand

$$F(s) = \frac{x^{s+1}}{s(s+1)} \left(-\frac{\zeta'(s)}{\zeta(s)} \right).$$

First we deform the vertical line from $c - i\infty$ to $c + i\infty$ to the path $\gamma(T)$ shown in Figure 2. (The segments of $\gamma(T)$ on the line $\mathrm{Re}(s) = 1$ consist of $T \leq t < \infty$, and $-\infty < t \leq -T$.) Here $T \geq 3$, and T will be chosen appropriately large later.

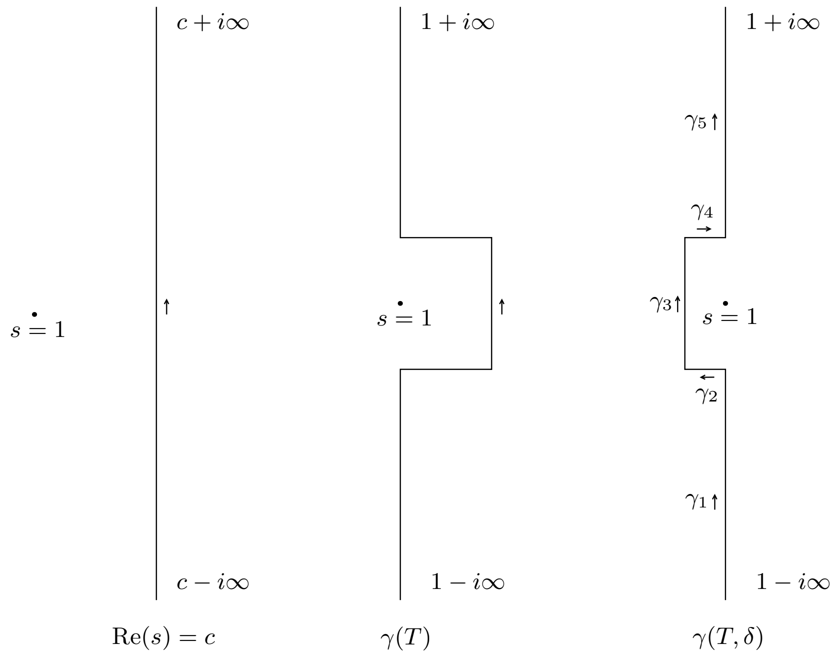

Figure 2. Three stages: the line $\mathrm{Re}(s) = c$, the contours $\gamma(T)$ and $\gamma(T, \delta)$

The usual and familiar arguments using Cauchy's theorem allow us to see that

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)\, ds = \frac{1}{2\pi i} \int_{\gamma(T)} F(s)\, ds. \tag{7}$$

Indeed, we know on the basis of Proposition 2.7 in Chapter 6 and Proposition 1.6 that $|\zeta'(s)/\zeta(s)| \leq A|t|^\eta$ for any fixed $\eta > 0$, whenever $s = \sigma + it$, $\sigma \geq 1$, and $|t| \geq 1$. Thus $|F(s)| \leq A'|t|^{-2+\eta}$ in the two (infinite) rectangles bounded by the line $(c - i\infty, c + i\infty)$ and $\gamma(T)$. Since F is regular in that region, and its decrease at infinity is rapid enough, the assertion (7) is established.

Next, we pass from the contour $\gamma(T)$ to the contour $\gamma(T, \delta)$. (Again, see Figure 2.) For fixed T, we choose $\delta > 0$ small enough so that ζ has no zeros in the box

$$\{s = \sigma + it,\ 1 - \delta \leq \sigma \leq 1,\ |t| \leq T\}.$$

Such a choice can be made since ζ does not vanish on the line $\sigma = 1$.

Now $F(s)$ has a simple pole at $s = 1$. In fact, by Corollary 2.6 in Chapter 6, we know that $\zeta(s) = 1/(s - 1) + H(s)$, where $H(s)$ is regular near $s = 1$. Hence $-\zeta'(s)/\zeta(s) = 1/(s - 1) + h(s)$, where $h(s)$ is holomorphic near $s = 1$, and so the residue of $F(s)$ at $s = 1$ equals $x^2/2$. As a result

$$\frac{1}{2\pi i}\int_{\gamma(T)} F(s)\, ds = \frac{x^2}{2} + \frac{1}{2\pi i}\int_{\gamma(T,\delta)} \frac{x^{s+1}}{s(s+1)} F(s)\, ds.$$

We now decompose the contour $\gamma(T, \delta)$ as $\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5$ and estimate each of the integrals $\int_{\gamma_j} F(s)\, ds$, $j = 1, 2, 3, 4, 5$, with the γ_j as in Figure 2.

First we contend that there exists T so large that

$$\left|\int_{\gamma_1} F(s)\, ds\right| \leq \frac{\epsilon}{2}x^2 \quad \text{and} \quad \left|\int_{\gamma_5} F(s)\, ds\right| \leq \frac{\epsilon}{2}x^2.$$

To see this, we first note that for $s \in \gamma_1$ one has

$$|x^{1+s}| = x^{1+\sigma} = x^2.$$

Then, by Proposition 1.6 we have, for example, that $|\zeta'(s)/\zeta(s)| \leq A|t|^{1/2}$, so

$$\left|\int_{\gamma_1} F(s)\, ds\right| \leq Cx^2 \int_T^\infty \frac{|t|^{1/2}}{t^2}\, dt.$$

Since the integral converges, we can make the right-hand side $\leq \epsilon x^2/2$ upon taking T sufficiently large. The argument for the integral over γ_5 is the same.

Having now fixed T, we choose δ appropriately small. On γ_3, note that

$$|x^{1+s}| = x^{1+1-\delta} = x^{2-\delta}\,,$$

from which we conclude that there exists a constant C_T (dependent on T) such that

$$\left|\int_{\gamma_3} F(s)\,ds\right| \le C_T x^{2-\delta}\,.$$

Finally, on the small horizontal segment γ_2 (and similarly on γ_4), we can estimate the integral as follows:

$$\left|\int_{\gamma_2} F(s)\,ds\right| \le C_T' \int_{1-\delta}^{1} x^{1+\sigma}\,d\sigma \le C_T'\frac{x^2}{\log x}\,.$$

We conclude that there exist constants C_T and C_T' (possibly different from the ones above) such that

$$\left|\psi_1(x) - \frac{x^2}{2}\right| \le \epsilon x^2 + C_T x^{2-\delta} + C_T'\frac{x^2}{\log x}.$$

Dividing through by $x^2/2$, we see that

$$\left|\frac{2\psi_1(x)}{x^2} - 1\right| \le 2\epsilon + 2C_T x^{-\delta} + 2C_T'\frac{1}{\log x},$$

and therefore, for all large x we have

$$\left|\frac{2\psi_1(x)}{x^2} - 1\right| \le 4\epsilon.$$

This concludes the proof that

$$\psi_1(x) \sim x^2/2 \quad \text{ as } x \to \infty\,,$$

and thus, we have also completed the proof of the prime number theorem.

Note on interchanging double sums

We prove the following facts about the interchange of infinite sums: if $\{a_{k\ell}\}_{1\le k,\ell<\infty}$ is a sequence of complex numbers indexed by $\mathbb{N}\times\mathbb{N}$, such that

$$\sum_{k=1}^{\infty}\left(\sum_{\ell=1}^{\infty}|a_{k\ell}|\right) < \infty, \tag{8}$$